

Detailed Answer Key

1A–D CONDUCTOR CONCEPTS

Concepts needed: electrostatic equilibrium; free-charge motion; conductor equipotential; surface charge; shielding

1. Assume a nonzero electric field remains inside the metal. A mobile charge would feel $F=qE$ and continue to move, so equilibrium is impossible.
2. Therefore the rearranged charges create a canceling field and the net interior field is zero. A Gaussian surface wholly inside the metal then encloses zero net charge.
3. The conductor is an equipotential: a tangential potential difference would drive charge along the surface. Thus the field just outside is normal to the surface.
4. Excess charge resides on the outer surface unless a cavity contains charge. Near a sharp point, surface charge density is larger and field lines crowd.
5. Shielding follows directly: induced surface charge cancels the static field in a closed conducting interior.

EQUATIONS AND CHECKS

$$\begin{aligned} E_{\text{metal}} &= 0 \\ E_{\text{outside}} &= \sigma/\epsilon_0 \end{aligned}$$

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1B–C CLEAN FIELD-LINE DRAWINGS

Concepts needed: field-line geometry; surface normal; surface charge density

1. Draw the conductor outline first and leave the interior blank; no field line should pass through the conducting material.
2. At every intersection with the surface, draw the line perpendicular to the local tangent. This enforces zero tangential field.
3. For a spherical conductor in a uniform field, use straight lines far away and smoothly bent lines around the sphere, with larger crowding on the sides where induced charge is concentrated.
4. For the irregular conductor, spread lines over broad regions and compress them near the pointed end. Lines never cross, and line density represents field magnitude.

EQUATIONS AND CHECKS

$$E_{\perp} = \sigma / \epsilon_0$$

field-line density $\propto |E|$

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2A NET FORCE ON +8 MC

Concepts needed: Coulomb law; one-dimensional sign convention; superposition

1. Set rightward force positive. The distance from q_1 to q_3 is $a+b=0.09$ m.
2. Because q_1 and q_3 are both positive, q_1 repels q_3 to the right: $F_{31} = k|q_1q_3|/(a+b)^2 = 35.5$ N.
3. Because q_2 is negative and q_3 positive, q_2 attracts q_3 to the left: $F_{32} = k|q_2q_3|/b^2 = 88.1$ N.
4. Add signed components: $F_3 = +35.5 - 88.1 = -52.6$ N. The negative sign means 52.6 N left.

EQUATIONS AND CHECKS

$$F = k|q_iq_j|/r^2$$

$$F_{\text{net}} = \Sigma F_i$$

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2B–C SYMBOLIC FIELDS AND BAND DIAGRAMS

Concepts needed: vector superposition; scalar potential; atomic levels; band formation

1. At $P=(a,b)$, the source-to-point distances are $r_1=\sqrt{a^2+b^2}$, $r_2=a$, and $r_3=b$.
2. Resolve each electric field into x and y components. The diagonal charge contributes to both components; the left and bottom charges contribute along one axis each.
3. Add components to obtain E_p . A charge q_4 at P experiences $F_4=q_4E_p$, so a negative q_4 reverses the field direction.
4. Potential is scalar: add kq/r for each source without component resolution.
5. For the drawing, show discrete levels for one atom, splitting into many nearby levels as atoms approach, then broad valence and conduction bands separated by a gap in the crystal.

EQUATIONS AND CHECKS

$$E_p = [kq_1 a / (a^2 + b^2)^{3/2} + kq_2 / a^2]i + [kq_1 b / (a^2 + b^2)^{3/2} + kq_3 / b^2]j$$

$$V_p = kq_1 / \sqrt{a^2 + b^2} + kq_2 / a + kq_3 / b$$

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3A–B PARALLEL PLATES AND NONUNIFORM SPHERE

Concepts needed: Gauss law; planar symmetry; Newton's second law; kinematics; spherical-shell integration

1. For plate area $A=L^2$, symmetry gives $E=Q/(\epsilon_0 A)=Q/(\epsilon_0 L^2)$. The particle's vertical force is qE , so $a_y=qQ/(\epsilon_0 mL^2)$.
2. The horizontal crossing time is $t=L/v$. With zero initial vertical velocity, $dy=\frac{1}{2}a_y t^2=qQ/(2\epsilon_0 mv^2)$.
3. For the sphere, a shell at radius r' has $dV=4\pi r'^2 dr'$. Substitute $\rho(r')=\rho_0(r'/R)^2$ to get $dq=4\pi\rho_0 r'^4 dr'/R^2$.
4. Integrate from 0 to R : $Q=(4\pi\rho_0/R^2)[r'^5/5]_0^R=4\pi\rho_0 R^3/5$. The radial dependence is why multiplying one density by total volume would be wrong.

EQUATIONS AND CHECKS

$$E = Q/(\epsilon_0 L^2); a_y = qQ/(\epsilon_0 mL^2); dy = qQ/(2\epsilon_0 mv^2)$$

$$Q = \int \rho dV = 4\pi\rho_0 R^3/5$$

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4A–D MASS-SPRING OSCILLATOR

Concepts needed: Hooke law; mechanical energy conservation; turning points; graph interpretation

1. At maximum displacement $x=A$, speed is zero and all mechanical energy is spring potential:

$$E = \frac{1}{2}kA^2 = \frac{1}{2}(100)(0.40)^2 = 8.0 \text{ J.}$$

2. At equilibrium $x=0$, spring potential is zero, so $\frac{1}{2}mv_{\text{max}}^2 = 8.0$. Thus $v_{\text{max}} = \sqrt{(16/3.3)} = 2.20 \text{ m/s}$.

3. At $x=0.03 \text{ m}$, solve $\frac{1}{2}mv^2 + \frac{1}{2}kx^2 = E$. This gives $v = \sqrt{[(16 - 100(0.03)^2)/3.3]} = 2.20 \text{ m/s}$.

4. Plot $U_s = \frac{1}{2}kx^2$ upward and $K = E - U_s$ downward. Both meet at $x = \pm A$; K is largest and U_s smallest at $x=0$.

EQUATIONS AND CHECKS

$$E = \frac{1}{2}kA^2 = 8.0 \text{ J}$$

$$v(x) = \sqrt{[(k/m)(A^2 - x^2)]} = 2.20 \text{ m/s}$$

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5 SOUND-PRESSURE LEVEL

Concepts needed: rms pressure; intensity; logarithmic decibels; spherical spreading

1. Define sound-pressure level relative to the reference pressure $p_0 = 20 \mu\text{Pa}$: $L_p = 20 \log_{10}(p_{\text{rms}}/p_0)$ dB.
2. For a change, subtract the two levels so the reference cancels: $\Delta L_p = 20 \log_{10}(p_2/p_1)$.
3. If $p_2 = 2p_1$, $\Delta L_p = 20 \log_{10} 2 = +6.02$ dB. If $p_2 = \frac{1}{2}p_1$, $\Delta L_p = 20 \log_{10}(\frac{1}{2}) = -6.02$ dB.
4. The result is not ± 2 dB because intensity scales as pressure squared: doubling pressure quadruples intensity, and $10 \log_{10} 4 = 6.02$.
5. The worksheet's spherical-wave context also gives $I = P/(4\pi r^2)$, so intensity falls as $1/r^2$ when no energy is absorbed.

EQUATIONS AND CHECKS

$$L_p = 20 \log_{10}(p_{\text{rms}}/p_0) \text{ dB}$$

$$I \propto p_{\text{rms}}^2; I_2/I_1 = (p_2/p_1)^2$$

$$I_2 = I_1(r_1/r_2)^2$$

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6 BOHR MODEL, IONS, AND MOLECULES

Concepts needed: Bohr model; Coulomb force; centripetal acceleration; quantized states; ionization; valence electrons; electron clouds; molecular formulas

1. Draw the nucleus and allowed electron states. The classical force balance for hydrogen is an attraction toward the nucleus equal in magnitude to the required centripetal force.
2. The balance is $k e^2/r^2 = m_e v^2/r$. It determines the speed for a chosen circular radius, while the quantum model adds the rule that only certain states are allowed.
3. Ionization requires enough energy to move an electron from a bound state to an unbound state. The remaining atom has a deficit of negative charge and is positive.
4. Sodium has one valence electron and commonly loses it to form Na^+ . Chlorine is one electron short of a filled outer shell and commonly gains one to form Cl^- .
5. An electron cloud shows probability rather than a fixed track. H_2 is diatomic; H_2O is triatomic. In multi-electron atoms, nucleus attraction and electron repulsion must be vector-added.

EQUATIONS AND CHECKS

$$k e^2/r^2 = m_e v^2/r$$

ionization: bound electron + energy $\rightarrow$ free electron + positive ion

molecular subscript = number of atoms

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7 POLARIZATION AND DISPLACEMENT VECTOR

Concepts needed: electric dipole; polarization density; susceptibility; permittivity; free and bound charge

1. For charges $\pm q$ separated by vector d , define the dipole moment $p=qd$, directed from negative to positive charge. This is the microscopic building block.
2. Sum all dipoles in volume V and divide by volume: $P=(1/V)\sum p_i$. Aligned dipoles give a macroscopic polarization vector; random orientations can cancel.
3. Start with the definition $D=\epsilon_0 E+P$. For a linear isotropic dielectric, the material response is $P=\epsilon_0 \chi_e E$, with dimensionless susceptibility χ_e .
4. Substitute and factor: $D=\epsilon_0 E+\epsilon_0 \chi_e E=\epsilon_0 (1+\chi_e)E$. Define $\epsilon=\epsilon_0 (1+\chi_e)$, so $D=\epsilon E$.
5. Draw E , P , and D collinear for the isotropic case, with dipoles aligned from negative toward positive. Keep E as the physical field and P as material response.

EQUATIONS AND CHECKS

$$\begin{aligned}
 p &= qd \\
 P &= (1/V) \sum p_i \\
 D &= \epsilon_0 E + P \\
 P &= \epsilon_0 \chi_e E \\
 D &= \epsilon_0 (1 + \chi_e) E = \epsilon E
 \end{aligned}$$

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8 IONS AND MOLECULAR POTENTIAL ENERGY

Concepts needed: potential energy; point-charge model; force as negative slope; Lennard–Jones terms; equilibrium stability

1. For point charges, write $U(r)=kq_1q_2/r$ with $U=0$ at infinite separation. Like charges give positive U ; opposite charges give negative U under this convention.

2. Use $F(r)=-dU/dr$. Read the graph by its slope: a negative slope corresponds to outward positive radial force, a positive slope to inward force, and a horizontal minimum to $F=0$.

3. Sketch a steep positive short-range wall, a negative attractive well at intermediate r , and an approach toward $U=0$ as r becomes large. Label the minimum as the equilibrium separation.

4. The Lennard–Jones expression combines repulsion and attraction: C_{12}/r^{12} dominates at short range, while $-C_6/r^6$ supplies longer-range attraction.

5. Na^+ and Cl^- contain nuclei and electrons, so they are composite. A point-charge approximation is useful outside the charge distribution, but a realistic short-range model needs additional repulsive physics.

EQUATIONS AND CHECKS

$$U=kq_1q_2/r$$

$$F(r)=-dU/dr$$

$$U_{\text{LJ}}=C_{12}/r^{12}-C_6/r^6$$

$$\text{at equilibrium: } dU/dr=0$$